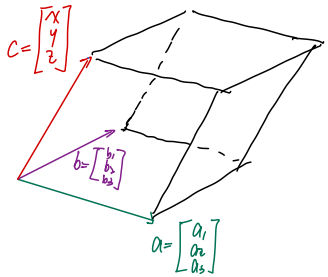
· Cross product

fixed b, a



signed volume $\left(\square\right) = f\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right)$

$$= \det\begin{bmatrix} x & a_1 & b_1 \\ y & a_2 & b_2 \\ z & a_3 & b_3 \end{bmatrix}$$

$$= x(a_2b_3 - a_3b_2) + y(a_3b_1 - a_1b_3) + z(a_1b_2 - a_2b_1)$$

(if $a, b, \begin{bmatrix}x\\y\\z\end{bmatrix}$ follow right hand rule, the sign is +, other wise its −)

f is linear in $\begin{bmatrix}x\\y\\z\end{bmatrix}$, thus can be expressed as inner produce

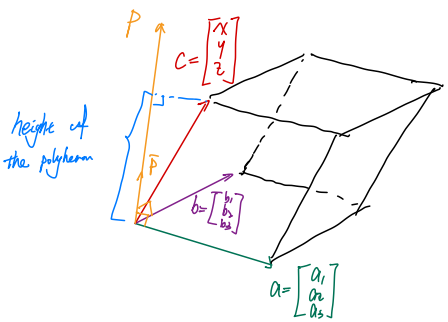
let $\bar{P}$ be a unit vector s.t. $\begin{cases}\bar{P}\perp a & \bar{P}\perp b \\ a\ b\ \bar{P} \text{ follows right-hand rule}\end{cases}$

Signed volume $\left(\square\right)$ = area $\left(\overset{b}{\underset{a}{\square}}\right) \cdot c^T\bar{P}$

$$= c^T\left(\bar{P}\cdot \text{area}\left(\overset{b}{\underset{a}{\square}}\right)\right)$$

$$= c^T P$$

$$= \begin{bmatrix}x\\y\\z\end{bmatrix}\begin{bmatrix}p_1\\p_2\\p_3\end{bmatrix}$$



P is called the cross-produce of a, b, written as $a\times b$

$P = a\times b$ is a vector s.t. $\begin{cases} P\perp a & P\perp b \\ a\ b\ p \text{ follows right-hand rule} \\ \|P\|_2 = \text{area}\left(\overset{b}{\underset{a}{\square}}\right) = \|a\|_2\|b\|_2\cdot\sin\theta \\ P\perp c \text{ gives the signed volume of } \overset{c}{\underset{a}{\overset{b}{\square}}} \end{cases}$

$$P = \begin{bmatrix}p_1\\p_2\\p_3\end{bmatrix} = \begin{bmatrix}a_2b_3 - a_3b_2\\a_3b_1 - a_1b_3\\a_1b_2 - a_2b_1\end{bmatrix} = \det\begin{bmatrix} i & a_1 & b_1 \\ j & a_2 & b_2 \\ k & a_3 & b_3 \end{bmatrix}$$

Cross-produce only defined for $\mathbb{R}^3$

In $\mathbb{R}^2$, no vector is orthogonal to 2 independent vectors

In $\mathbb{R}^4$, inf num of vector are orthogonal to 2 independent vectors

$$p = a \times b = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

can be written as $p = \begin{cases} [a]_\times b = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} | \\ b \\ | \end{bmatrix} \\ [b]_\times^\top a = \begin{bmatrix} 0 & b_3 & -b_2 \\ -b_3 & 0 & b_1 \\ b_2 & -b_1 & 0 \end{bmatrix} \begin{bmatrix} | \\ a \\ | \end{bmatrix} \end{cases}$

Theorem ☆

$\forall$ rotation matrix $R \in SO(3)$, $w \in \mathbb{R}^3$

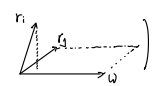
we have $R[w]_\times R^\top = [Rw]_\times$

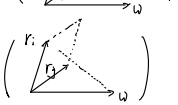
let $R = \begin{bmatrix} — r_1^\top — \\ — r_2^\top — \\ — r_3^\top — \end{bmatrix}$

$$R[w]R^\top = R[w]\begin{bmatrix} | & | & | \\ r_1 & r_2 & r_3 \\ | & | & | \end{bmatrix}$$

$$= \begin{bmatrix} — r_1^\top — \\ — r_2^\top — \\ — r_3^\top — \end{bmatrix} \begin{bmatrix} | & | & | \\ w \times r_1 & w \times r_2 & w \times r_3 \\ | & | & | \end{bmatrix}$$

if $i = j$ $(w \times r_i) \perp r_i$, thus $r_i^\top (w \times r_i) = 0$

if $i \neq j$ $r_i^\top (w \times r_j) =$ volume $\left(\right.$  $\left. \right)$

$=$ volume $\left(\right.$  $\left. \right)$

$$= w^\top (r_j \times r_i) = \pm w^\top r_k$$

$i=1, j=2$ $\quad w^\top (r_2 \times r_1) = -w^\top r_3$

$i=1, j=3$ $\quad w^\top (r_3 \times r_1) = w^\top r_2$

$i=2, j=3$ $\quad w^\top (r_3 \times r_2) = -w^\top r_1$

Since R is a rotation matrix

$r_1 \times r_2 = r_3$

$r_2 \times r_3 = r_1$

$r_3 \times r_1 = r_2$

thus $R[w]R^\top = \begin{bmatrix} 0 & -w^\top r_3 & w^\top r_2 \\ w^\top r_3 & 0 & -w^\top r_1 \\ -w^\top r_2 & w^\top r_1 & 0 \end{bmatrix} = \begin{bmatrix} w^\top r_1 \\ w^\top r_2 \\ w^\top r_3 \end{bmatrix}_\times = [Rw]_\times$

Matrix Exponential & Rotation

Theorem
$$\begin{cases} e^{-A} = (e^{A})^{-1} \\ e^{A+B} = e^{A}e^{B} \quad \text{if } AB=BA \\ e^{A}\cdot e^{-A} = I \end{cases}$$

Theorem: if A is skew-symmetric, e^{A} is orthonormal

$-A = A^{T} \Rightarrow e^{-A} = e^{A^{T}} \Rightarrow (e^{A})^{-1} = (e^{A})^{T} \Rightarrow e^{A}$ is orthonormal

Theorem: if $w = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $[w]_{\times} = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}$

then $e^{[w]_{\times}}$ is a rotation matrix about w by $\|w\|_2$ radius

and $\dot{u} = [w]_{\times}u$ is a rotation about w with angular velocity $\|w\|_2$ rad/s

$$\det\left([w]-\lambda I\right) = \det\begin{bmatrix} -\lambda & -z & y \\ z & -\lambda & -x \\ -y & x & -\lambda \end{bmatrix} = -\lambda^3 - xyz + xyz - \lambda x^2 - \lambda y^2 - \lambda z^2$$
$$= -\lambda(\lambda^2 + x^2 + y^2 + z^2)$$

$\lambda = \begin{bmatrix} 0 \\ \pm j\cdot\|w\|_2 \end{bmatrix}$ Also $\underbrace{\begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}}_{A}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$, thus $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is a eigenvector of A with eigenvalue 0

$AA = -A^{T}A$ is a symmetric matrix, thus it has orthogonal real eigenvectors

Suppose A has eigenvector $V_r \pm jV_c$

then $A(V_r + jV_c) = j\|w\|_2(V_r + jV_c) = \|w\|_2(-V_c + jV_r)$

$AA(V_r + jV_c) = \|w\|_2 \cdot j \cdot (AV_r + jAV_c)$

$= \|w\|_2 \cdot j \cdot \|w\|_2(-V_c + jV_r)$

$= -\|w\|_2^2(V_r + jV_c)$

thus V_r, V_c are eigenvectors of AA, with eigenvalue $-\|w\|_2^2$

thus V_r, V_c and $[x,y,z]^{T}$ are orthogonal ($[x,y,z]^{T}$ is one of A's eigenvectors)

$A(V_r + jV_c) = j\|w\|_2(V_r + jV_c) = \|w\|_2(-V_c + jV_r)$

$AV_r = -\|w\|V_c \qquad AV_c = \|w\|V_r$

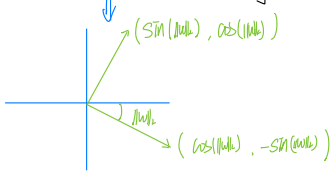
$V_r^{T}AV_c = V_r^{T}\|w\|V_r = \|w\|\|V_r\|_2^2$

$V_c^{T}A^{T}V_r = -V_c^{T}AV_r = -V_c^{T}\cdot -\|w\|V_c = \|w\|\|V_c\|_2^2$

$\Big\} \|V_r\|_2 = \|V_c\|_2$ WLOG, can assume $\|V_r\| = \|V_c\| = 1$

$e^{[w]}$ has eigenvalue $\begin{cases} e^0 = 1 & \text{with eigenvector } w = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \\ e^{\pm \|w\| i} = \cos(\|w\|) \pm i \sin(\|w\|) & \text{with eigenvector } Vr \pm iVc \end{cases}$

$$e^{[w]} = \begin{bmatrix} | & | & | \\ Vr & Vc & w/\|w\| \\ | & | & | \end{bmatrix} \begin{bmatrix} \cos(\|w\|) & \sin(\|w\|) & \\ -\sin(\|w\|) & \cos(\|w\|) & \\ & & 1 \end{bmatrix} \begin{bmatrix} | & | & | \\ Vr & Vc & w/\|w\| \\ | & | & | \end{bmatrix}^{-1}$$



remark for theorem ☆

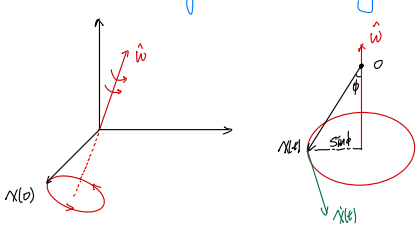
from theorem $R[w]R^T = [Rw]$, it can be well-interpreted by $e^{[Rw]}$

$e^{[Rw]} = \exp m(R[w]R^T) = R \exp m([w])R^T$

both $e^{[Rw]}$ (in base coord) and $Re^{[w]}R^T$ (in R coord) rotates about Rw by $\|Rw\| = \|w\|$ radius

· Frame Rotation

consider rotating about $\hat{w}$ with angular velocity $\dot{\theta}$ rad/s



$\|\dot{x}(t)\|_2 = \dot{\theta} \sin\phi$ magnitude

$\frac{\dot{x}(t)}{\|\dot{x}(t)\|_2} = \frac{\hat{w} \times x(t)}{\|\hat{w} \times x(t)\|_2} = \frac{\hat{w} \times x(t)}{\sin\phi}$ direction

$\dot{x}(t) = \frac{\hat{w} \times x(t)}{\sin\phi} \cdot \dot{\theta} \sin\phi = \hat{w}\dot{\theta} \times x(t)$

$x(t) = \exp m([\hat{w}\theta] \cdot t)\, x(0)$

$$\begin{aligned} \exp([\hat{w}]\theta) &= I + \theta[\hat{w}] + \frac{\theta^2}{2!}[\hat{w}]^2 + \frac{\theta^3}{3!}[\hat{w}]^3 + \frac{\theta^4}{4!}[\hat{w}]^4 \\ &= I + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} \cdots\right)[\hat{w}] + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} \cdots\right)[\hat{w}]^2 \\ &= I + \sin\theta\,[\hat{w}] + (1 - \cos\theta)[\hat{w}]^2 \end{aligned}$$

$[\hat{w}]^4 = -[\hat{w}]^2$
$[\hat{w}]^5 = -[\hat{w}]^3 = [\hat{w}]$
} easily seen since $[\hat{w}]$ has eigenvalue $\{0, \pm i\}$

rotate $\{B\}$ about $\hat{w}$ with angular velocity $\dot{\theta}$

let $B(t)$ be the moving body frame

I be the fixed inertial frame

$R_{IB}(t)$ be the representation of $B(t)$ in I

$\hat{w}_{/I}$ and $\hat{w}_{/B(t)}$ be the representation of $\hat{w}$ in I and $B(t)$

let $w_{/I} = \hat{w}_{/I} \cdot \dot{\theta}$ $\quad R_{IB}(t) = \begin{bmatrix} | & | & | \\ r_1(t) & r_2(t) & r_3(t) \\ | & | & | \end{bmatrix}$

$$\dot{R}_{IB}(t) = \begin{bmatrix} | & | & | \\ w_{/I} \times r_1(t) & w_{/I} \times r_2(t) & w_{/I} \times r_3(t) \\ | & | & | \end{bmatrix} = [w_{/I}]\, R_{IB}(t)$$

$$\dot{R}_{IB}(t) = [w_{/I}]\, R_{IB}(t) \quad \Rightarrow \quad \dot{R}_{IB}(t)\, R_{IB}(t)^{-1} = [w_{/I}]$$

Angular velocity in $\{I\}$

$$w_{/I} = R_{IB}(t)\, w_{/B(t)}$$

$$[w_{/B(t)}] = [R_{IB}(t)^{-1} w_{/I}] = R_{IB}(t)^{-1} [w_{/I}]\, R_{IB}(t)$$

$$= R_{IB}(t)^{-1} \dot{R}_{IB}(t)\, R_{IB}(t)^{-1}\, R_{IB}(t)$$

$$= R_{IB}(t)^{-1} \dot{R}_{IB}(t)$$

$$\dot{R}_{IB}(t) = R_{IB}(t)\, [w_{/B(t)}]$$

Angular velocity in $B(t)$

- Quaternion

1. quaternion in terms of axis-angle vector

$$q = \begin{bmatrix} \cos(\frac{\theta}{2}) \\ r \sin(\frac{\theta}{2}) \end{bmatrix} \quad \begin{matrix} \} \text{ scalar part} \\ \} \text{ vector part} \end{matrix}$$

2. $q^T q = 1$

rotations are unit quaternions, easy to normalize

3. q and $-q$ represent the same rotation

$$-\begin{bmatrix} \cos(\frac{\theta}{2}) \\ r \sin(\frac{\theta}{2}) \end{bmatrix} = \underbrace{\begin{bmatrix} \cos(\frac{2\pi-\theta}{2}) \\ -r \sin(\frac{2\pi-\theta}{2}) \end{bmatrix}}_{\text{rotate } 2\pi-\theta \text{ about } -r}$$

4. Quaternion multiplication

$$\begin{bmatrix} s_1 \\ v_1 \end{bmatrix} * \begin{bmatrix} s_2 \\ v_2 \end{bmatrix} = (s_1 + a_1 i + b_1 j + c_1 k)(s_2 + a_2 i + b_2 j + c_2 k)$$

$$\begin{cases} jk = -kj = i \\ ki = -ik = j \\ ij = -ji = k \\ i^2 + j^2 + k^2 = -1 \end{cases}$$ same as cross product

$$= s_1 s_2 + s_1(a_2 i + b_2 j + c_2 k) + s_2(a_1 i + b_1 j + c_1 k) + \underbrace{(a_1 i + b_1 j + c_1 k)(a_2 i + b_2 j + c_2 k)}_{-(a_1 a_2 + b_1 b_2 + c_1 c_2) + \text{cross terms}}$$

$$q_1 * q_2 = \begin{bmatrix} s_1 \\ v_1 \end{bmatrix} * \begin{bmatrix} s_2 \\ v_2 \end{bmatrix} = \begin{bmatrix} s_1 s_2 - v_1^T v_2 \\ s_1 v_2 + s_2 v_1 + v_1 \times v_2 \end{bmatrix} = \begin{cases} \left[\begin{array}{c|c} s_1 & -v_1^T \\ \hline v_1 & s_1 I + [v_1]_\times \end{array}\right] \begin{bmatrix} s_2 \\ v_2 \end{bmatrix} = L(q_1)\, q_2 \\ \left[\begin{array}{c|c} s_2 & -v_2^T \\ \hline v_2 & s_2 I - [v_2]_\times \end{array}\right] \begin{bmatrix} s_1 \\ v_1 \end{bmatrix} = R(q_2)\, q_1 \end{cases}$$

5. Quaternion Identity $\begin{bmatrix} \cos\theta \\ r\sin(\theta) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

6. Quaternion Conjugate: $\begin{bmatrix} s \\ v \end{bmatrix}^{-1} = \begin{bmatrix} s \\ -v \end{bmatrix}$

$$\left.\begin{aligned} L(q)^T &= L(q^{-1}) = L(q)^{-1} \\ R(q)^T &= R(q^{-1}) = R(q)^{-1} \end{aligned}\right\}$$ just like $Q^T = Q^{-1}$ for rotation matrix

$$\left.\begin{aligned} (q_1 * q_2)^{-1} &= q_2^{-1} * q_1^{-1} \\ q * q^{-1} &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned}\right\}$$ just like matrix

$$\begin{aligned} q_2^{-1} * q_1^{-1} * q &= L(q_2^{-1}) L(q_1^{-1})\, q \\ &= L(q_2)^T L(q_1)^T q \\ &= \left(L(q_1) L(q_2)\right)^T q \\ &= L(q_1 * q_2)^T q \\ &= L(q_1 * q_2)^{-1} q \end{aligned}$$

7. rotate a vector

$$\begin{bmatrix} 0 \\ x_{rt} \end{bmatrix} = q_{rt} * \begin{bmatrix} 0 \\ x_{rb} \end{bmatrix} * q_{rt}^{-1} = \begin{cases} L(q)\, R(q^{-1}) \begin{bmatrix} 0 \\ x_{rb} \end{bmatrix} = L(q) R(q)^T \begin{bmatrix} 0 \\ x_{rb} \end{bmatrix} \\ R(q^{-1}) L(q) \begin{bmatrix} 0 \\ x_{rb} \end{bmatrix} = R(q)^T L(q) \begin{bmatrix} 0 \\ x_{rb} \end{bmatrix} \end{cases}$$

8. Quaternion kinematics

look at small increment in rotation $\delta q = \begin{bmatrix} \cos(\frac{\theta}{2}) \\ r\sin(\frac{\theta}{2}) \end{bmatrix}$ with small θ

$$\delta q * (q * \begin{bmatrix} 0 \\ x_{rb} \end{bmatrix} * q^{-1}) * \delta q^{-1} = (\delta q * q) * \begin{bmatrix} 0 \\ x_{rb} \end{bmatrix} * (\delta q * q)^{-1}$$

$$\delta q * q = \begin{bmatrix} \cos(\frac{\theta}{2}) \\ \hat{w}\sin(\frac{\theta}{2}) \end{bmatrix} * q \approx \begin{bmatrix} 1 \\ 0 \end{bmatrix} * q + \begin{bmatrix} 0 \\ \frac{1}{2} w\theta \end{bmatrix} * q = q + \begin{bmatrix} 0 \\ \frac{1}{2} w\alpha t \end{bmatrix} * q$$

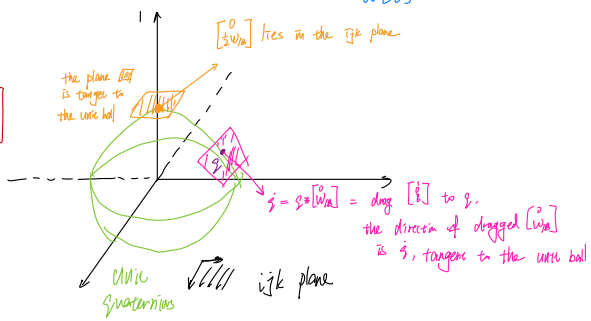
$$\frac{\delta q * q - q}{\delta t} \approx \begin{bmatrix} 0 \\ \frac{1}{2} w \end{bmatrix} * q$$

$$\dot{q}_{I/I} = \begin{bmatrix} 0 \\ \frac{1}{2}\omega_{I/B} \end{bmatrix} * q_{I/I} = q_{I/I} * \begin{bmatrix} 0 \\ \frac{1}{2}\omega_{I/B} \end{bmatrix} * q_{I/I}^{-1} * q_{I/I} = q_{I/I} * \begin{bmatrix} 0 \\ \frac{1}{2}\omega_{I/B} \end{bmatrix}$$

① from body frame perspective

$$\begin{cases} \dot{Q} = Q[\omega_{I/B}] \\ \dot{q}_{I/I} = q_{I/I} * \begin{bmatrix} 0 \\ \frac{1}{2}\omega_{I/B} \end{bmatrix} \end{cases}$$

$$Q[\omega_{I/B}]\begin{bmatrix} 0 \\ 0 \end{bmatrix} = Q[\omega_{I/B}]\,\hat{b}_{1/B} = Q\,\dot{\hat{b}}_{1/B} = \dot{\hat{b}}_{1/I} = \dot{Q}\begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



② from Inertial frame perspective

$$\begin{cases} \dot{Q} = [\omega_{I/I}]\,Q \\ \dot{q}_{I/I} = \begin{bmatrix} 0 \\ \frac{1}{2}\omega_{I/I} \end{bmatrix} * q_{I/I} \end{cases}$$

